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ORIGINAL RESEARCH ARTICLE

Stress Corrosion Cracking of AA7075 Aluminum Alloy Processed via Additive Friction Stir Deposition

Ebenezer Acquah, Shubhrodev Bhowmik, A.F.M.Monowar Hossain, and Nilesh Kumar

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High-performance structures are needed for many structural applications, including land, air, and marine-based transportation systems. To that end, additive friction stir deposition (AFSD), a solid-state manufacturing process that operates on the principles of friction stir processing, is quite a promising additive manufacturing tool with the potential to create metallic materials with desired properties. This study focuses on understanding the stress corrosion cracking (SCC) susceptibility of AA7075 aluminum alloy processed via AFSD. The SCC was evaluated by carrying out a slow strain rate tensile (SSRT) testing in 3.5 wt.% NaCl solution at room temperature in as-processed and post-processing heat treatment (T73-type temper) conditions. The tensile testing of the alloy in as-processed was done at initial strain rates (constant crosshead velocity) of 10^{-4}, 10^{-5}, and 10^{-6} s^{-1}, whereas the alloy in T73 condition was subjected to SSRT testing under similar conditions at 10^{-6} s^{-1}. Scanning electron microscopy, electron backscattered diffraction, and transmission electron microscopy were used for microstructural characterization. The grain size obtained after deposition was 3.81 ± 2 µm. The SSRT testing results show that the as-processed AA7075 aluminum alloy exhibited a high susceptibility to SCC at 10^{-6} s^{-1}. The T73 heat treatment caused reduced susceptibility of the AFSD-processed AA7075. The TEM results showed the secondary phases were rich in Cu in the as-processed case. The level of Cu enrichment in coarse secondary phases was significantly smaller after T73 heat treatment. The high potential difference between the matrix and the precipitate due to Cu depletion in the matrix might have made the as-deposited alloy more susceptible to SCC. The T73 heat treatment was found to be effective in improving the SCC resistance of the AFSD-processed AA7075 alloy.

Keywords additive manufacturing, aerospace, aluminum, fractography, mechanical testing, nonferrous metals

1. Introduction

Additive manufacturing (AM) is a transformative technology enabling the manufacture of near-net shape objects through a layer-by-layer build-up process. Unlike traditional manufacturing, AM relies on 3D model data, eliminating the need for physical tooling (Ref 1). This approach has experienced rapid adoption, driven by the demand for high-performance components in diverse sectors such as energy, aerospace, automotive, construction, and medicine (Ref 2). The American Society for Testing and Materials (ASTM) categorizes AM processes into seven groups: directed energy deposition (DED), material extrusion, binder jetting, powder bed fusion (PBF), vat photopolymerization, material jetting, and sheet lamination (Ref 3, 4). The layer bonding technique and consolidation mechanism mostly determine each AM technology's capabilities, difficulties, and final material properties (Ref 5). The primary approaches for fabricating metals and alloys in AM are fusion-based methods and solid-state additive manufacturing processes (Ref 5, 6). These methods represent two paradigms in AM technology, each offering unique advantages and applications within the broader additive manufacturing landscape.

In the past decade, solid-state additive manufacturing has witnessed significant advancements (Ref 7). According to Tuncer and Bose, this field can be categorized into two main methods based on material bonding: sinter-based solid-state AM and mechanical deformation-based techniques (Ref 5). Sinter-based AM encompasses binder jetting (Ref 8), screen printing (Ref 9) and metallic extrusion (Ref 10), while mechanical deformation-based techniques include ultrasonic additive manufacturing (UAM) (Ref 11), cold spray (Ref 12), and additive friction stir processes (Ref 13). Among these methods, additive friction stir deposition (AFSD) stands out for its ability to subject feedstock rods to severe global plastic deformation at elevated temperatures. This process induces dynamic recrystallization, resulting in a uniform, equiaxed, and fine microstructure (Ref 4).

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Ebenezer Acquah, Shubhrodev Bhowmik, A.F.M. Monowar Hossain, and **Nilesh Kumar**, Department of Metallurgical and Materials Engineering, The University of Alabama, Tuscaloosa, AL 35487, USA. Contact e-mail: nkumar@eng.ua.edu.

Research on AFSD encompasses various aspects including equipment development, material flow characteristics, microstructure evolution, and changes in mechanical properties (Ref 14). Melike and Yahya reviewed studies on the microstructure and mechanical properties of steel, titanium, nickel, magnesium alloys, and aluminum alloys deposited using AFSD (Ref 13). AFSD holds particular significance for aluminum alloys among these metals due to their thermal sensitivity and weldability issues in fusion-based methods. AFSD offers advantages such as low residual stresses and high-strength defect-free part production (Ref 13). Anderson-Wedge et al. (Ref 15), Rutherford et al. (Ref 16), and Avery et al. (Ref 17) have investigated the fatigue behavior of as-deposited AA2219, AA6061, and AA7075 aluminum alloys, respectively. Jordon et al. (Ref 18) examined AFSD-deposited AA5083 sourced from machine chips, while Beck et al. (Ref 19) investigated the mechanical properties of AA5083 feed material derived from rods and chips. The effect of corrosion on AA5083 manufactured via AFSD was investigated by Li et al. (Ref 20). Rivera et al. (Ref 21) explored how the mechanical behavior of AFSD AA2219 is influenced by grain refinement and texture.

Yoder et al. (Ref 22) studied the mechanical properties of hyper-eutectic Al-Si automotive machining chips deposited via AFSD, demonstrating improved mechanical properties compared to conventional cast aluminum chips. Additionally, Li et al. (Ref 23) investigated the mechanical properties of 2195 Al-Li alloy manufactured by the AFSD method. Chen et al. (Ref 24) explored the relationship between the microstructure and mechanical properties of as-deposited AA6061. At the same time, Phillips et al. (Ref 25) highlighted the influence of deposition paths and material flow at the interface on microstructure and tensile behavior. Additionally, the effects of post-processing treatments like solid solution strengthening and aging on the mechanical performance of AFSD AA6061 have been studied by Beck et al. (Ref 26). Ghadimi et al. (Ref 27) and Chen et al. (Ref 28) have also revealed the sensitivity of microstructure and mechanical properties of AA6061 deposition to AFSD process parameters such as layer thickness, rotational speed, and feed rate. Babaniaris et al. (Ref 29) determined the effect of the AFSD process on precipitation in the as-deposited AA6063 swarf. Notably, there is a gap in research regarding the impact of stress corrosion cracking (SCC) phenomena on the mechanical properties of AFSD-processed aluminum alloys.

Other additive manufacturing techniques encounter issues like microstructural anisotropy, the presence of pores, high residual stresses, and oxides, all of which contribute to susceptibility to SCC (Ref 5, 30, 31). In addition, research has not yet explored the SCC behavior of materials processed by UAM, while existing cold spray studies mainly focus on SCC behavior in coated substrates rather than bulk materials. The aluminum alloy AA7075, which belongs to the 7XXX series, finds extensive application in the automotive, aerospace, and defense sectors due to its exceptional mechanical strength-to-weight ratio (Ref 32). Achieving equiaxed grains in AA7075 has posed challenges for certain additive manufacturing methods. Su et al. noted that the microstructure of AA7075 aluminum alloy processed through laser additive manufacturing is linked to cracking and the formation of columnar grains (Ref 33). Although AA7075 aluminum alloy has been processed using additive friction stir deposition, showing desired microstructure and mechanical properties, it is important to understand the SCC behavior of the alloy in AFSD-processed conditions (Ref 17, 34).

This research focuses on understanding the effects of SCC on AA7075 processed via AFSD. In this work, a multi-layer build of AA7075 is fabricated with AFSD and subjected to SCC in as-processed and under T73 temper conditions using quasi-static strain rates in salt solution at room temperature. Microstructural characterization is conducted with a scanning electron microscope (SEM), electron backscatter diffraction (EBSD), and transmission electron microscope (TEM). The resulting mechanical properties and microstructure are analyzed to develop insight into the impact of SCC on AFSD-processed AA7075.

2. Materials and Methods

2.1 Feedstock and Substrate for AFSD: AA7075-T6

The material used in this experiment was a commercially available AA7075-T6 aluminum plate with a nominal chemical composition shown in Table 1. A feedstock rod with a cross-section of 9.5 × 9.5 mm^2 and 304.8 mm length and a substrate of cross-section 9.5 (T) × 76.2 (W) mm^2 and 152.4 mm length used for AFSD were machined out from 304.8 × 304.8 mm^2 and 19.05 mm thick AA7075-T6 plate using a state-of-the-art Mitsubishi Electric wire-electrical discharge machine (EDM), which has insignificant effect on the microstructure or surface properties of the aluminum alloy as it produces no measurable heat-affected zone (HAZ) or base-metal cracking (Ref 35).

2.2 Additive Friction Stir Deposition Process

The feedstock rod and substrate of AA7075-T6 alloy was prepared using wire-EDM. B8 MELD machine was used during the AFSD. The face of the AFSD tool had four "teardrop" protrusions adjacent to the feedstock rod channel (38.1 mm diameter). The protrusions were to aid in heat generation and mechanical mixing at the substrate/feedstock interface (Fig. 1a). The substrate was clamped onto the print bed. The feedstock rods were coated with graphite lubricant before loading into the hollow rotating tool. Initially, the rotating tool with the loaded feedstock rod was lowered until the tool protrusions partly plunged into the surface of the substrate. The AFSD tool with the feedstock rod was spun at a rotational rate (Ω) of 225 rpm. The actuator then pushed the feedstock rod out of the hollow tool while rotating at a 25.4 mm/min feed rate. During the deposition process, the feedstock material softens and spreads between the rotating tool head and substrate due to the combination of frictional heating and compressive force from the actuator. A G-code was run to provide uniform and controlled motion of the tool head and deposit upon yielding. Other parameters used for deposition were a feed rate of 68.85 mm/min, a tool traverse speed of 125 mm/min, and a 95.25 mm-long travel (center-to-center of the rotating tool) along the longitudinal direction of the substrate. A 50.8 mm build was constructed by layer-by-layer deposition (26 layers), with each layer being close to 2 mm in thickness. A schematic of the AFSD build is shown in Fig. 1(b).

Table 1 Nominal chemical composition of AA7075-T6 from the supplier McMaster-Carr®

	Si	Fe	Cu	Mn	Mg	Cr	Zn	Ti	Ni	Zr	Al
wt.%	0-0.4	0-0.5	1.2-2.0	0-0.3	2.1-2.9	0.18-0.287	5.1-6.1	0-0.2	<0.01	<0.05	Bal.

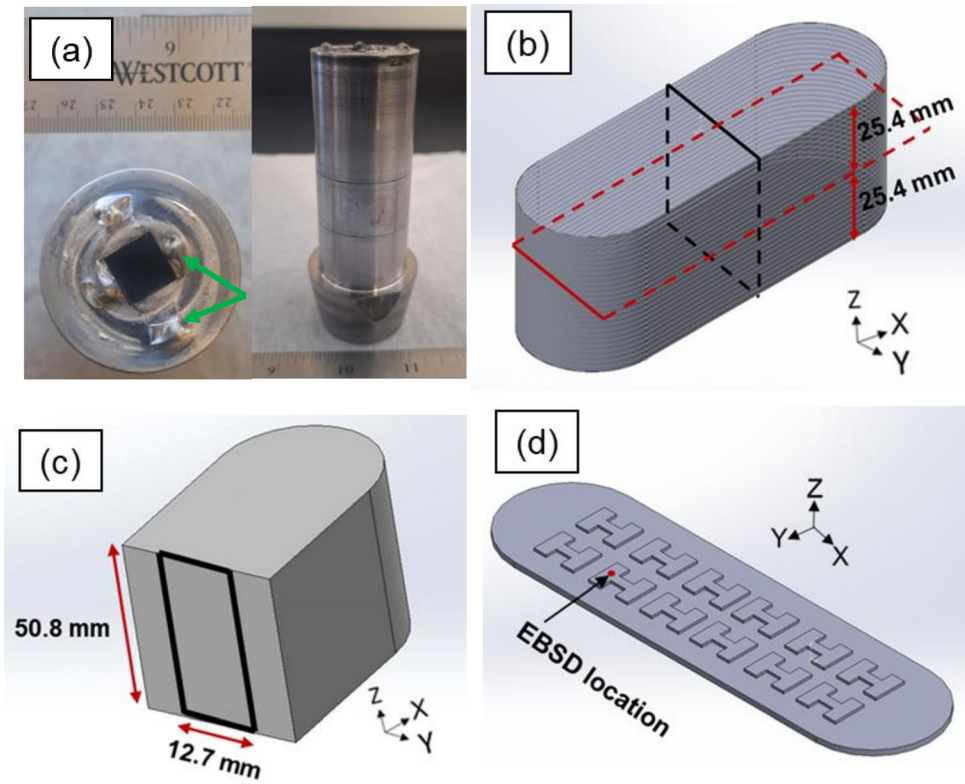

Fig. 1 (a) Rotating tool with four teardrop protrusions (green arrows): top view and side view, (b) Schematic of build: black dotted line showing the sectioning and red dotted line showing the sectioning of the build into two halves, (c) Half of the build: black rectangle indicating the area for hardness measurement, and (d) The first layer below the red sectioning plane (at 25.4 mm depth from top) indicating how the tensile test coupons were machined out. X, Y, and Z represent processing, transverse, and build directions, respectively

2.3 Microstructural Characterization

The tensile specimen indicated with a red circle in Fig. 1(d) was used for EBSD analysis. The tensile specimen was cold mounted, followed by grinding and polishing with SiC-based grinding papers and diamond suspensions with different sizes. The tensile specimen was subjected to vibratory polishing with 0.03 µm colloidal silica for 3 h. The EBSD measurements were taken using Apreo SEM (Thermo Fisher Scientific). The measurements were conducted at 20 keV and 6.4 nA using the APEX system (AMETEK, Inc.) with 0.35 µm step size in a 50 × 50 μm^2 region at a magnification of 800X. The EBSD data was analyzed with the OIM-8 analysis package (AMETEK). A cleaning of less than 5% of unindexed points was done on the EBSD data.

TEM samples were prepared from both as-processed and T73 heat-treated (details provided in section 2.5) conditions of the AFSD AA7075 build. For TEM sample preparation, 3 mm diameter disks were punched out from 20 × 20 mm strips of the as-processed and T73 condition samples, each approximately 120 µm thick. These disks were then mechanically ground to ~ 60 µm. Struers TenuPol-5 machine was used to perform twin-jet electropolishing on the TEM samples. The electropolishing solution consisted of 90 ml of distilled water, 730 ml of ethanol, 100 ml of ethylene glycol monobutyl ether, and 109 ml of 50% perchloric acid. The as-processed condition was performed at − 25 °C and 30 V, while the T73 condition was performed at − 24 °C and 22 V. The TEM analysis of as-processed and T73 conditions was performed using Talos F200i TEM.

Following the SSRT test of the as-processed and T73 heat-treated AFSD tensile specimens in the salt solution (3.5 wt.%) and air environment, respectively at the 10^{-6} s^{-1} initial strain rates, the fractured surface of the specimens was observed under the Apreo SEM (Thermo Fisher Scientific) at 20 keV, a 1.6 nA, and a dwell time of 10 µs.

2.4 Mechanical Properties

2.4.1 Microhardness Measurement. Vickers microindentation hardness testing was performed with an automated hardness tester (Clemex) to investigate the hardness distribution of the build. A cross-section of 3 mm thickness perpendicular

to the longitudinal direction was taken halfway from the build machined using wire-EDM and is indicated by the black dash lines in Fig. 1(b). A 12.7 × 50.8 mm area was sectioned from the 3 mm thick sample, as shown in (Fig. 1c). The sample was cold mounted, ground, and polished down to 0.05 µm surface finish before microhardness analysis. An array of (10 × 10) indents was made. Indent spacing was set to 1 and 2 mm in the horizontal and vertical directions, respectively. A single indent load of 300 g and a dwell time of 10 s were used during the hardness measurement.

2.4.2 Stress Corrosion Cracking (SCC) Testing. The SCC of the as-processed AFSD AA7075 was investigated via slow strain rate tensile (SSRT) testing. Although SCC initiation typically occurs under constant load in real-world applications, the SSRT test is widely accepted for assessing SCC susceptibility in metallic materials (Ref 36). The longitudinal section of the AFSD AA7075 build was separated into two parts, denoted by the red dash lines in Fig. 1(b). The location of sectioning was guided by hardness measurement results. The bottom half of the build was considered for SCC studies for the reason explained later. The first layer was machined out using wire EDM. Dog bone-shaped tensile test coupons of gauge dimensions 6.45 (L) × 2.95 (W) mm^2 were machined out from the sectioned layer such that the tensile loading axis was parallel to the longitudinal direction of the build as shown in Fig. 1(d). The tensile test coupons were polished to 0.05 µm (alumina colloidal suspension) with a final tensile test specimen thickness of ~1.0 mm. The tensile tests were evaluated using universal testing equipment (ADMET, Inc.) shown in Fig. 2(a). The tensile test was performed at room temperature in 3.5 wt.% NaCl solution (following ASTM G47-22) as well as in air at initial strain rates of 10^{-6} s^{-1}, 10^{-5} s^{-1}, and 10^{-4} s^{-1} corresponding to crosshead velocities of 0.0233 mm/h, 0.233 mm/h, and 2.33 mm/h, respectively. The choice of strain rate was based on the observation that most SSRT tests are conducted at 10^{-6} s^{-1}, 10^{-5} s^{-1}, and 10^{-4} s^{-1} strain rates (Ref 36). Similar SCC tests were also performed for the second layer from the build's bottom half to compare as-processed AFSD AA7075 and T73 heat-treated AFSD AA7075 coupons. Since AFSD-processed AA7075 showed the highest SCC susceptibility at a 10^{-6} s^{-1} strain rate, the SCC testing in T73 was conducted only at the strain rate of 10^{-6} s^{-1} to understand the impact of T73 heat treatment on the SCC susceptibility of the alloy. Figure 2(b) shows a close-up view of the tensile test set-up, showing a tensile test coupon mounted on a specimen holder submerged in a salt solution contained in a cylindrical salt solution bath.

2.5 Parameters for Heat Treatment (T73)

For studying SCC in T73 heat-treated conditions (shows better SCC resistance than the alloy in T6 temper condition), the tensile coupons machined out from the second layer of the bottom half part were first subjected to solution heat treatment at 475 °C for 2 h in a Thermolyne F48010 muffle furnace (Thermo Scientific), followed by rapid quenching in water at room temperature. It was followed by a two-step aging heat treatment: heating the specimen at 107 °C for 8 h and at 163 °C for 24 h with the same furnace. For brevity, T73 heat treatment will be referred to as T73 in subsequent discussions.

3. Results

3.1 AFSD of AA7075 Alloy and Microhardness Measurement

Figure 3(a) shows the side and isometric view of the AFSD-processed AA7075 aluminum alloy. It shows the height and length of the deposited build was ~50.8 mm and ~138.5 mm, respectively. Figure 3(b) displays the hardness distribution of

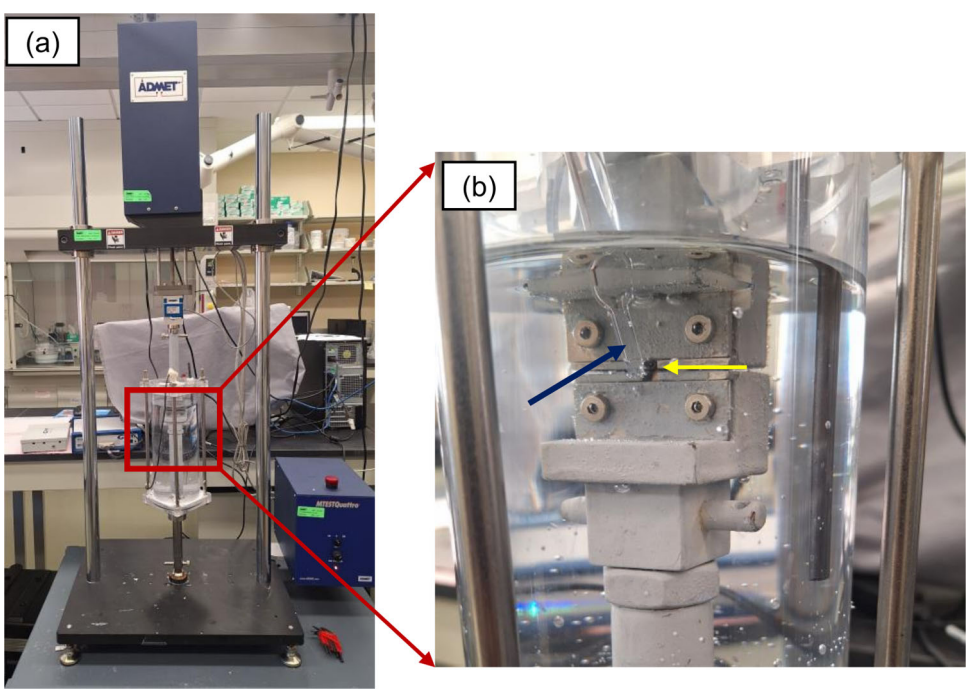

Fig. 2 (a) Experimental set-up to perform the slow strain rate tensile test, (b) magnified image of the set-up in 3.5 wt.% NaCl solution. The blue arrow shows the reference electrode to measure the open circuit potential, and the yellow arrow shows the location of the tensile specimen

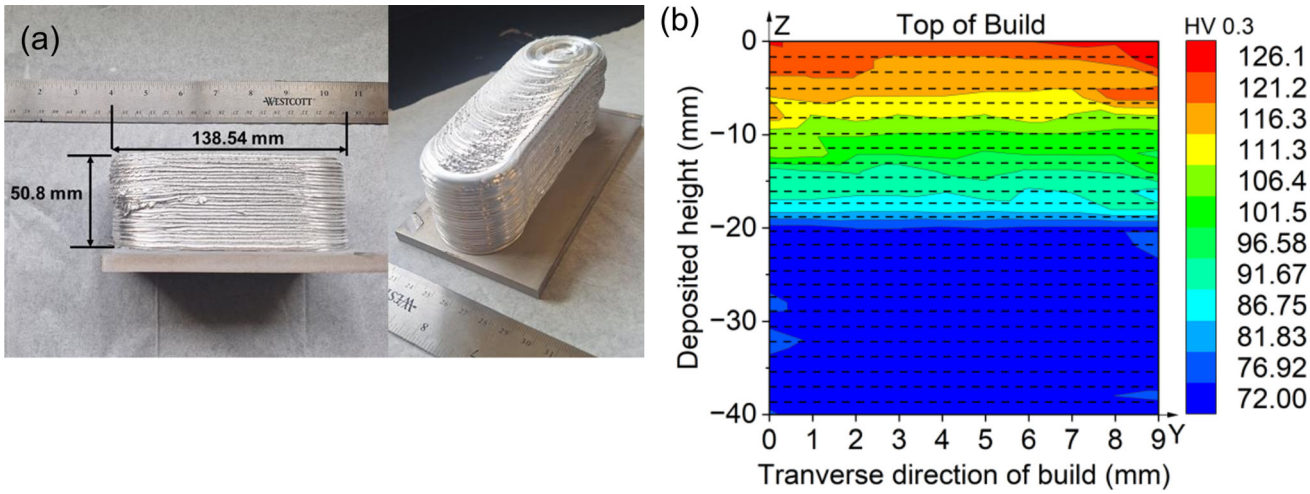

Fig. 3 (a) Build of AA7075 aluminum alloy AFSD. (b) Hardness distribution on the cross-section of the build

the as-deposited AA7075 build taken at the location shown in Fig. 1(c). The top half of the build (25.4 mm height from the top of the build) exhibited a hardness gradient, while the other 25.4 mm height (bottom half of the build) showed a uniform hardness distribution. The maximum hardness was found to be∼126 HV 0.3, by the top layer, and the minimum hardness was∼72 HV 0.3. Due to the uniformity of hardness in the bottom half of the build, the specimens for microstructural characterization and SCC study were machined out from the bottom half of the build (showing hardness of∼72 HV 0.3).

3.2 Microstructural Characterization

3.2.1 EBSD Analysis. The results of the EBSD analysis performed on the sample in as-processed conditions are shown in Fig. 4 and 5. The scanned area consisted of∼1030 grains. The inverse pole figure (IPF) map is shown in Fig. 4(a), superimposed with high angle grain boundaries (HAGBs) depicted as black boundaries and low angle grain boundaries (LAGBs) depicted as white boundaries. The image quality (IQ) map shown in Fig. 4(a_1) is superimposed with HAGBs (red) and LAGBs (green). In addition, the area fraction and cumulative fraction of grain size were investigated considering a grain boundary misorientation of 15°. This is shown in Fig. 4 (b). The average grain size was 3.81±2.00 μm. A very small fraction (<10%) of grains were in the ultrafine grain (<1 μm) region. The misorientation angle distribution, MAD (correlated and uncorrelated), superimposed with the McKenzie distribution (theoretical misorientation angle distribution for materials with a face-centered cubic crystal structure) is shown in Fig. 4 (c). The experimental data (correlated) shows a LAGB (2° and 15°) fraction of 0.16. The uncorrelated MAD curve closely follows the McKenzie distribution curve, suggesting the absence of any strong texture in the as-processed condition.

The aspect ratio (minor axis divided by major axis) is shown in Fig. 4(d) with an average of 0.64±0.2 μm, and most of the grains fall close to the ratio 1, meaning grains are predominantly equiaxed. Figure 5(a) and (b) show the pole figures (PF) and orientation distribution function (ODF) plot, respectively, of the as-processed AA7075. These indicate the presence of certain types of shear texture. Some ideal texture components are superimposed on the ODF plots. Some prominent components are $\{1\bar{1}\bar{1}\}\langle 110\rangle$, $\{\bar{1}11\}\langle\bar{1}\bar{1}0\rangle$ and $\{11\bar{1}\}\langle 112\rangle$. The first 2 are known as A and the last one is A* (Ref 37), respectively, and are generated because of shearing during processing. Weak Goss ($\{110\}\langle 001\rangle$) and Brass ($\{110\}\langle 1\bar{1}2\rangle$) components are also observed and could be because of recrystallization.

3.2.2 TEM Results. The bright field (BF) TEM and scanning transmission electron microscopy-high angle annular dark field (STEM-HAADF) were used (Fig. 6 and 7) to reveal the type of intermetallic particles and composition of the particles in the as-processed and T73 conditions of the AFSD AA7075. The intermetallic particles in the as-processed condition consisted of different particle sizes. This means the intermetallic particles are of different morphologies. A higher magnification micrograph taken from the region bound within the square box shown in Fig. 6(a) is shown in Fig. 6(b). Needle-shaped and disk-shaped structures are found within the grains, while the coarser intermetallic particles are mostly located on the grain boundaries, with a few within the grains. A wide precipitation-free zone (PFZ) can be observed. The STEM EDS map of the as-processed condition was taken from the area indicated by the yellow box in Fig. 6(a). The mapping results show that the precipitates are enriched with Cu, Zn, and Mg, which are shown in Fig. 6(d), (e) and (f), respectively.

The precipitates in the T73 condition are composed of numerous fine black spots and coarse precipitates, as shown in Fig. 7. The coarser precipitates are present on the grain boundaries and intragranular regions. Some elongated needle-shaped precipitates can also be seen along the grain boundaries. A higher magnification micrograph taken from the yellow square box is shown in Fig. 7(b). The green arrow lines show narrow PFZs in Fig. 7(a). The fine precipitates found in the T73 condition are distributed in the intragranular regions. The EDS map of the high magnification bright field image in Fig. 7(b) is shown in Fig. 7(c) to (f). The mapping results indicate that the coarser precipitates in the T73 condition are highly enriched with Zn and Mg with a significantly less amount of Cu than in coarser precipitates seen in as-processed condition. The fine precipitates in T73 condition are mostly enriched with Zn and Mg with a slightly elevated level of Cu enrichment compared with that in the matrix. These results indicate two different types of precipitates present in T73 heat-treated conditions.

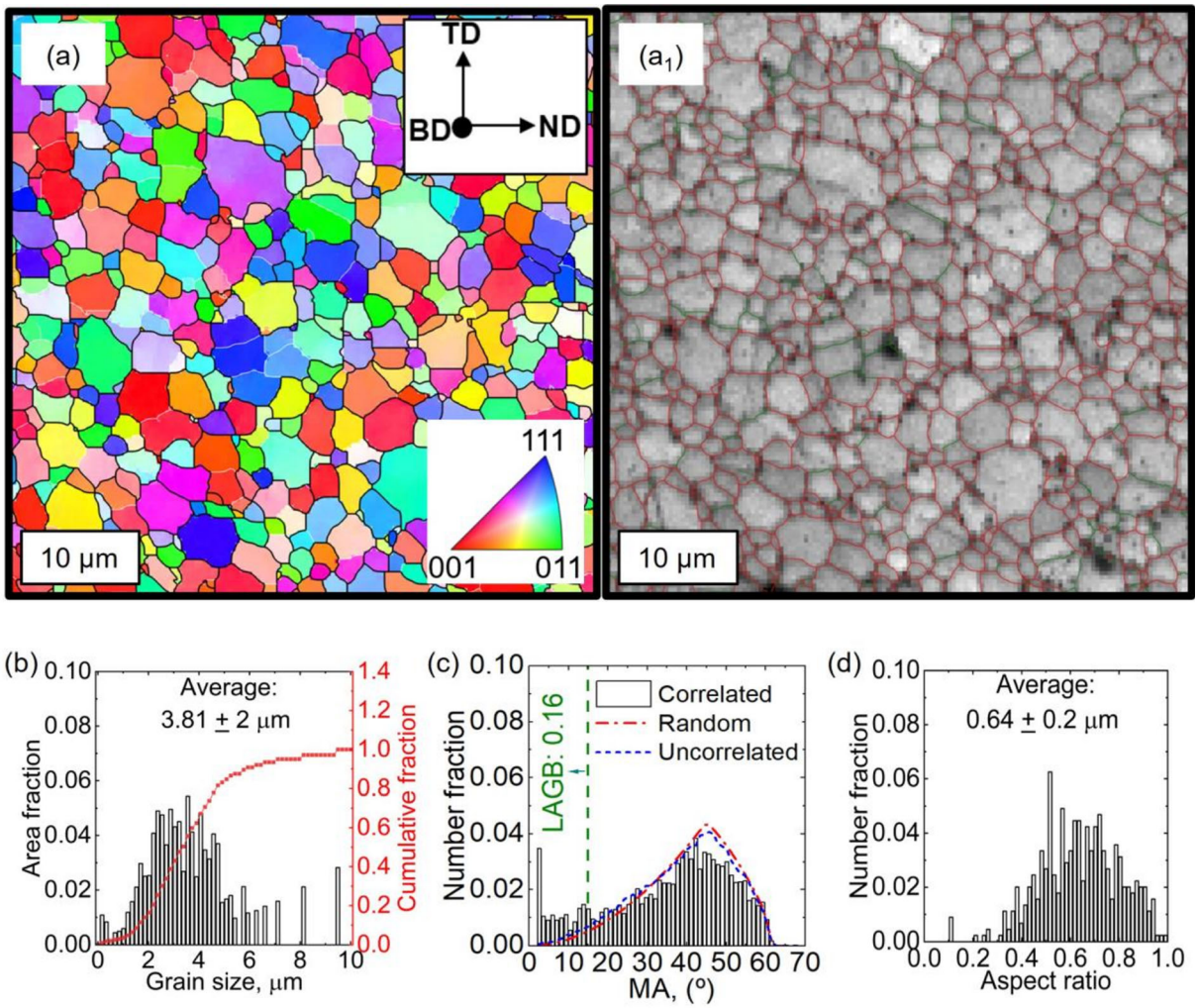

Fig. 4 EBSD results of as-processed AA7075; (a) an IPF map superimposed with LAGBs (2-15°) and HAGBs (15-180°). White and black lines were used to indicate LAGBs and HAGBs, respectively, (a1) shows the IQ map superimposed with LAGBs (2-15°) and HAGBs (15-180°). Green and red lines represent LAGBs and HAGBs, respectively, (b) grain size distribution plot, (c) misorientation angle distribution plot. MA: misorientation angle, (d) aspect ratio

3.3 Mechanical Properties and Fractographs

3.3.1 Stress Corrosion Cracking Characterization. To understand the SCC behavior of as-processed and T73 AFSD AA7075 conditions, dog-bone-shaped tensile coupons were tested at 10^{-6} s^{-1} (as-processed and heat-treated), 10^{-5} s^{-1} (as-processed) and 10^{-4} s^{-1} (as-processed) initial strain rates in salt environment and air conditions. Figure 8 shows the engineering stress vs. engineering strain curves for all the specimens tested in air and 3.5 wt.% NaCl solution for AFSD AA7075 (as-processed). Fig. 8(a) and (b) represent the raw and smoothened data of the engineering stress vs. engineering strain curves, respectively. Moreover, Fig. 8(b) plots stress as a function of the plastic part of the elongation. The curve data was smoothened for clarity of presentation. It was observed that the specimens tested in air showed a higher elongation (%) than their salt solution counterparts. The specimen tested at a 10^{-6} s^{-1} strain rate in the salt solution demonstrated a maximum decrease in elongation, followed by 10^{-5} s^{-1} and 10^{-4} s^{-1} strain rates.

Table 2 lists the yield strength (YS), ultimate tensile strength (UTS), and plastic part of the elongation to failure (%) obtained from the engineering stress vs. engineering strain of as-processed and T73 conditions. For the 10^{-6} s^{-1} initial strain rate tested specimen in a salt solution, the as-processed AA7075 exhibited a decrease in YS, UTS, and elongation compared to that observed in the air at the same initial strain rate. At 10^{-5} s^{-1} initial strain rate (as-processed), a decrease was also observed in YS, UTS, and a reduction in elongation compared to the as-processed air-tested specimen of the same initial strain rate. At 10^{-4} s^{-1} strain rate, although the YS remained the same compared to the air-tested specimen of the same initial strain rate, the UTS and the elongation to failure experienced a decrease. The T73 tensile specimen, when tested in air, showed an increase in YS (2.5 times) and UTS (~1.7 times) with a reduction in elongation to failure (~51%) compared to the as-processed air-tested tensile specimen at 10^{-6} s^{-1} initial strain rate. The T73 heat-treated condition tensile specimen at 10^{-6} s^{-1} initial strain rate in a salt environment saw an increase in YS (2 times) and UTS (~1.6 times) but a decrease in elongation (~40% reduction) compared to the YS, UTS, and elongation, respectively, of the as-processed counterpart tested under the same condition. The foregoing observation is summarized pictorially in Fig. 9(a), which shows the variation

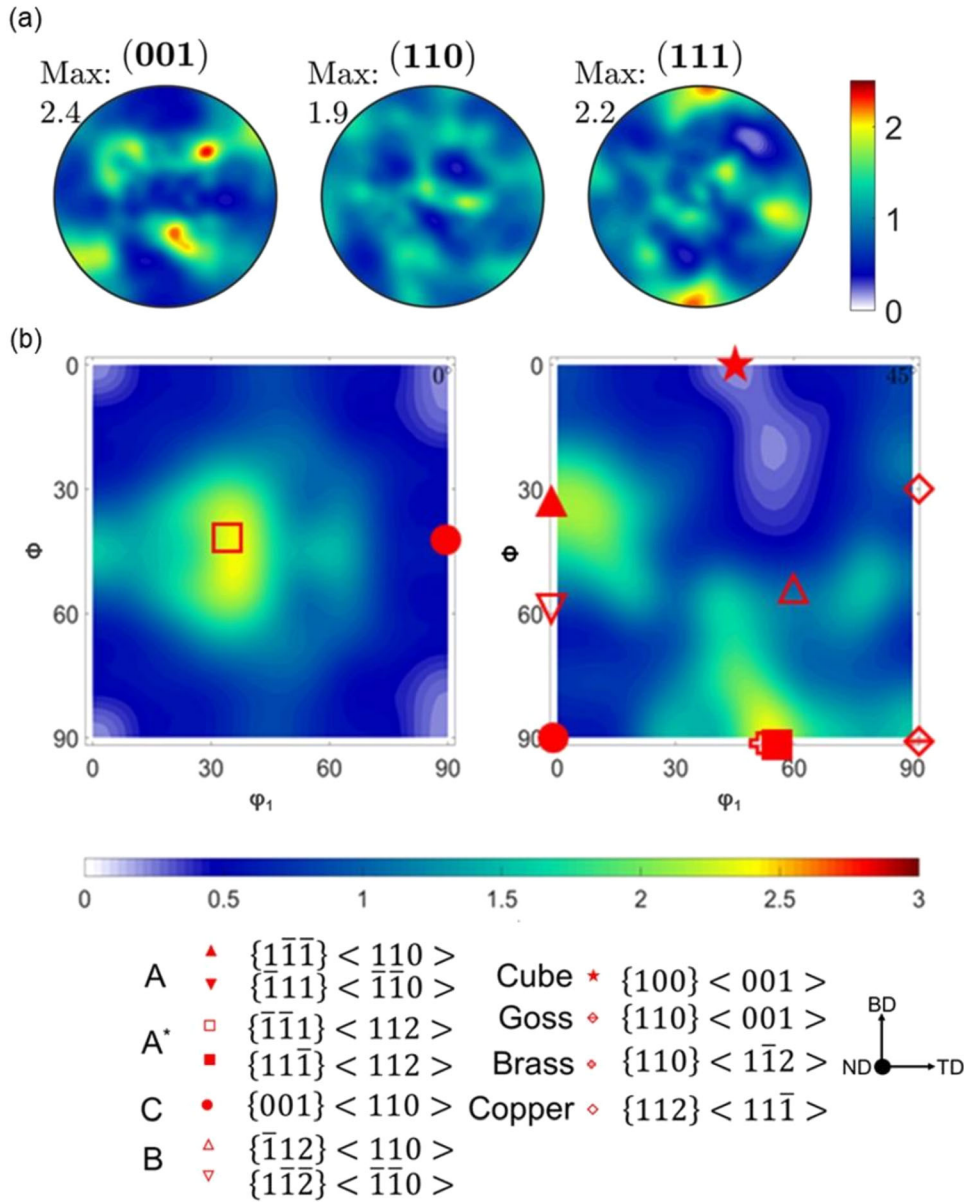

Fig. 5 Texture of the as-processed AA7075 from EBSD. (a) pole figures where the ND is out of plane. (b) orientation distribution function for at φ_2=0° and 45° section. ND, BD, TD represent the normal, build and transverse directions, respectively

of elongation (plastic part) as a function of initial strain rates tested in air and salt environments in as-processed and T73 conditions.

Figure 9(b) shows the SCC susceptibility index (I_{scc}) versus strain rate plot. From the plot, the SCC susceptibility is highest at 10^{-6} s^{-1} strain rate while 10^{-5} s^{-1}, and 10^{-4} s^{-1} showed almost the same susceptibilities to SCC. The high susceptibility index at 10^{-6} s^{-1} suggests that the susceptibility of as-processed AA7075 increased with a decrease in strain rate. The SCC susceptibility index was defined with the following equation (Ref 38, 39,

$$\text{SCC susceptibility index}_{\text{rate}} = \frac{\varepsilon_{\text{rate}}^{\text{air}} - \varepsilon_{\text{rate}}^{\text{sol}}}{\varepsilon_{\text{rate}}^{\text{air}}} \times 100$$

where $\varepsilon_{\text{rate}}^{\text{sol}}$ and $\varepsilon_{\text{rate}}^{\text{air}}$ are elongations in salt environment and air, respectively, at a given strain rate. Table 3 lists the I_{scc} for the three strain rates in the as-processed condition and for the T73 condition at 10^{-6} s^{-1}. For the 10^{-6} s^{-1} strain rate, SCC susceptibility is∼71%; and the sample tested at 10^{-5} s^{-1}, and 10^{-4} s^{-1} showed a susceptibility index of∼35% and∼38%, respectively. The susceptibility index for the T73 specimen at 10^{-6} s^{-1} strain rate was∼65%. This shows a decrease in SCC susceptibility compared to the as-processed condition at 10^{-6} s^{-1} strain rate.

3.3.2 Fractography. Figure 10(a) and (b) show low magnification SEM fractographs of the as-processed condition specimens tested at a 10^{-6} strain rate in air and salt environments, respectively. Similarly, Fig. 11(a) and (b) present the low magnification fracture surface images for the T73 specimens tested under the same conditions. Examining the various features visible in these fractographic images revealed that the fracture mode was dimpled features. The presence of dimples represents ductile fracture. However, the analysis showed that the relative proportion of these dimpled features

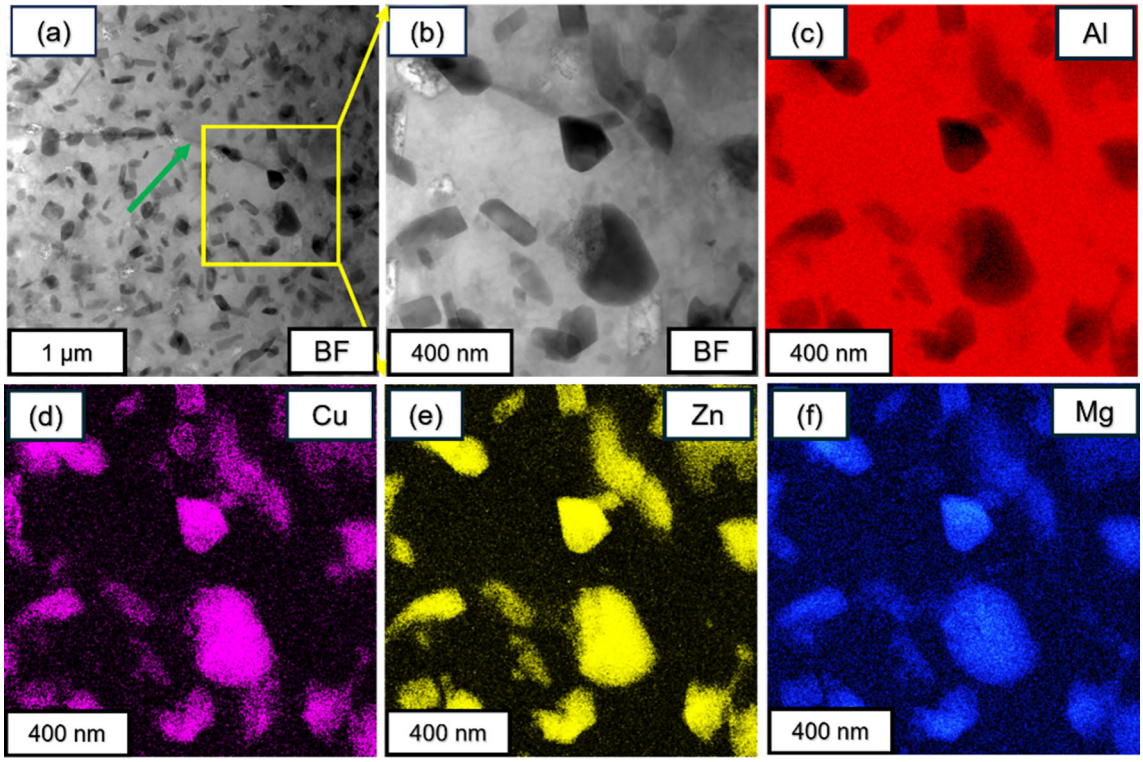

Fig. 6 TEM result of the as-processed condition of AFSD. (a) and (b) Bright-field image (BF) at low magnification, and high magnification (from the yellow box), respectively. (c) Al overlay of particles, (d) Cu overlay of particles, (e) Zn overlay of particles, (f) Mg overlay of particles. The green arrow shows a grain boundary known to be where SCC initiates in Al alloys. The presence of needle-shaped and disk-shaped structures within the grains while intermetallic particles are located on the grain boundary

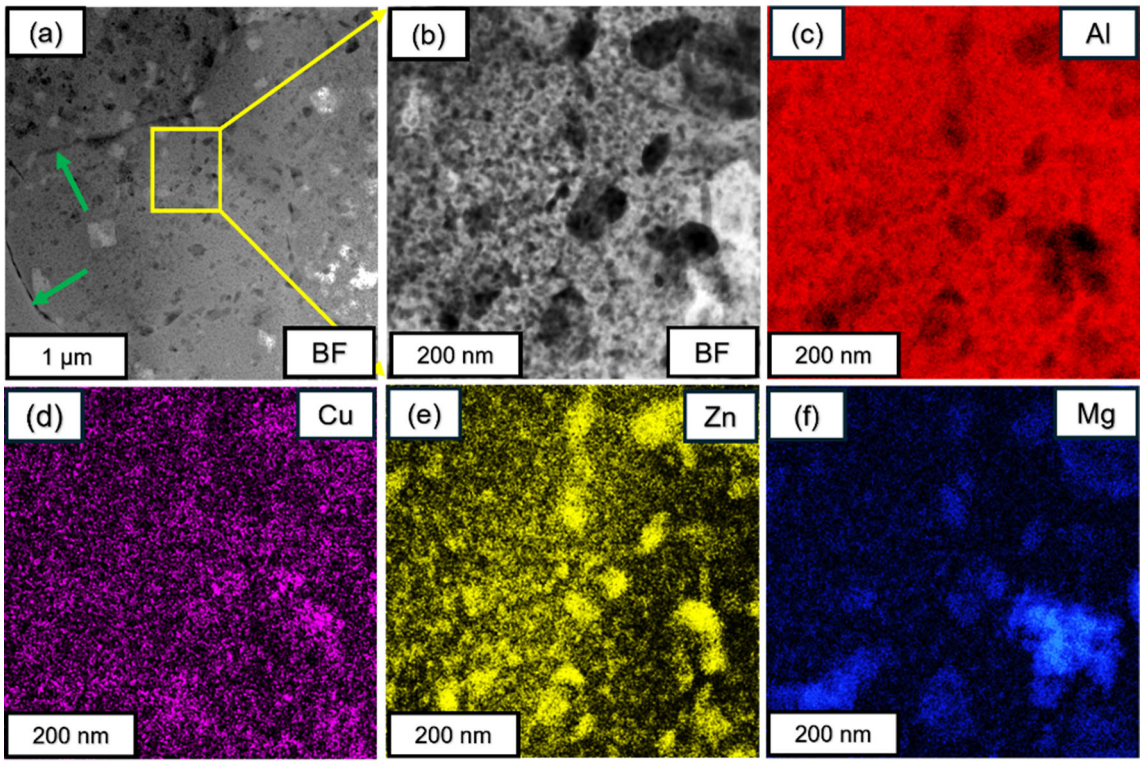

Fig. 7 TEM result of the T73 condition of AFSD. (a) and (b) Bright-field (BF) image at low magnification, and high magnification (from the yellow box), respectively. (c) Al overlay of particles, (d) Cu overlay of particles, (e) Zn overlay of particles, (f) Mg overlay of particles. Green arrows show the grain boundaries known to be where SCC initiates in Al alloys. Presence of GP(II) phase (numerous fine black spots precipitates within the grains) and coarser precipitates located at the grain boundaries and intragranular regions

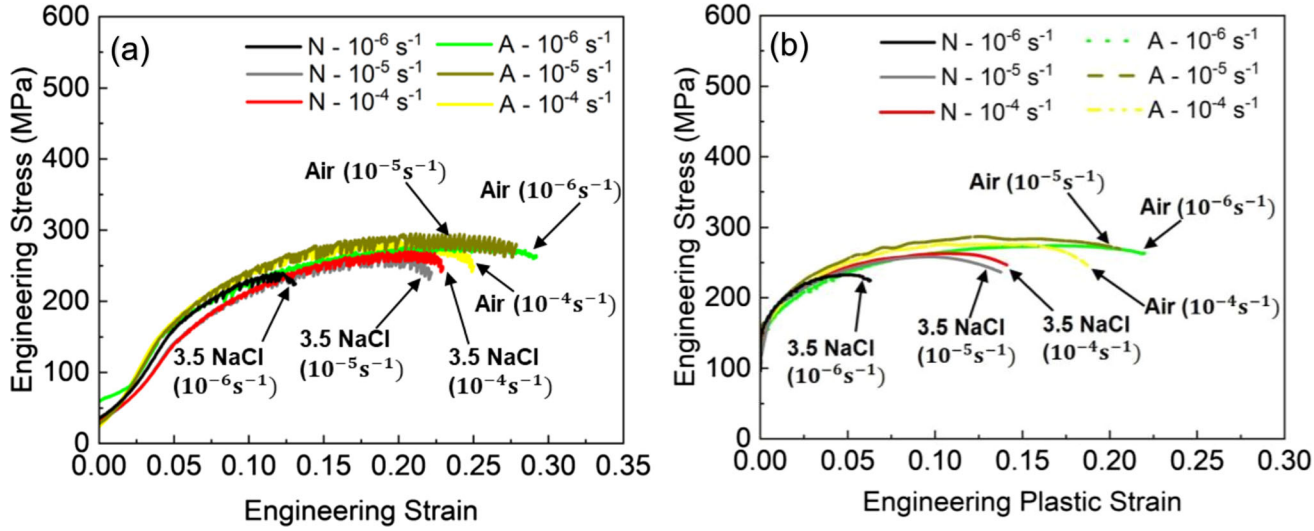

Fig. 8 Engineering stress—engineering strain curves of as-processed AFSD AA7075 in 3.5 wt.% NaCl and air. (a) raw data and (b) smoothened data with only plastic part of the elongation shown

Table 2 Mechanical properties for as-processed and T73 conditions of AFSD AA7075 tested in air and salt solution (3.5 wt.% NaCl) at room temperature

Condition	S. no.	Strain rate, s^{-1}	Environment	Yield strength, MPa	Ultimate tensile strength, MPa	Elongation, %
As-processed	1.	10^{-4}	Air	155±2	273±1	19.9±1.20
	2.		Salt	151±3	263±2	12.5±1.70
	3.	10^{-5}	Air	166±11	290±8	19.8±0.55
	4.		Salt	156±5	259±0	12.9±0.75
	5.	10^{-6}	Air	167±9	289±15	20.8±1.07
	6.		Salt	158±3	236±3	6.0±0.35
T73	7.	10^{-6}	Air	428±5	477±1	10.3±0.30
	8.		Salt	356±41	379±42	3.6±0.40

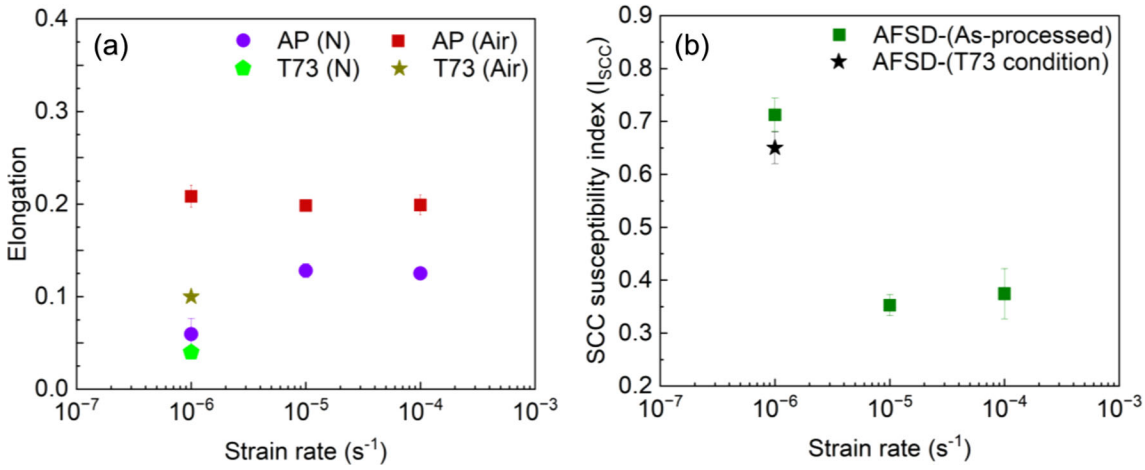

Fig. 9 (a) Elongation vs strain rate of as-processed (AP) and T73-condition of AFSD AA7075 in 3.5 wt.% NaCl (N) and air conditions and (b) SCC susceptibility index of as-processed and T73-condition of AFSD AA7075 in 3.5 wt.% NaCl

varied for the as-processed and T73 conditions and testing environment. A high fraction of dimpled features were seen on the as-processed air-tested condition (evident from a high magnification fractography shown in Fig. 10(a1) compared to the as-processed specimens tested in salt solution (Fig. 10b1). The same observation holds good for the T73 specimens tested in air and salt envirnoments. The comparison of fractographs corresponding to as-processed and T73 specimens tested in salt solution at 10^{-6} s^{-1} indicates that the T73 exhibited larger and deeper dimples, indicating higher ductility during SCC.

Table 3 Stress corrosion cracking susceptibility index for the as-processed and T73 conditions of AFSD AA7075

Condition	S. no.	Strain rate, s^{-1}	Stress corrosion cracking susceptibility index, I_{scc}
As-processed	1.	10^{-4}	0.375
	2.	10^{-5}	0.353
	3.	10^{-6}	0.713
T73	4.	10^{-6}	0.650

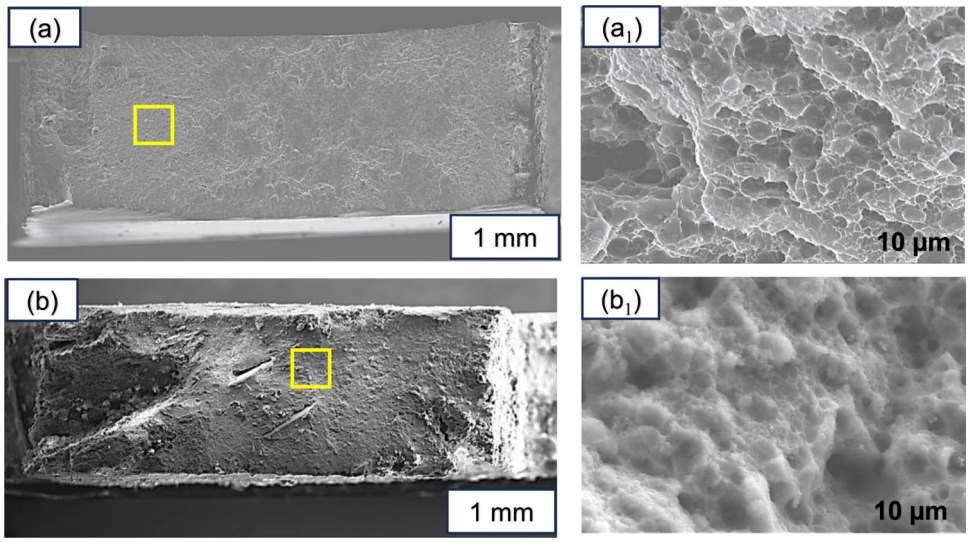

Fig. 10 Fractography of fractured tensile specimens tested in as-processed condition. (a) and (b) at low magnification in air and salt environments, respectively at 10^{-6} s^{-1} strain rate, (a_1) and (b_1) at high magnification in air and salt environments, respectively

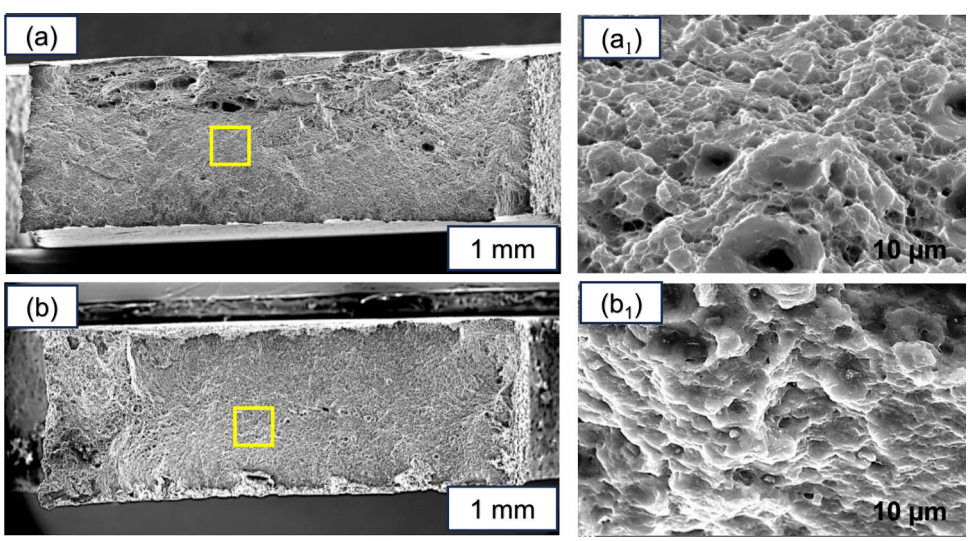

Fig. 11 Fractography of fractured tensile specimens tested in T73 heat treatment condition. (a) and (b) at low magnification in air and salt environments, respectively at 10^{-6} s^{-1} strain rate, (a_1) and (b_1) at high magnification in air and salt environments, respectively

4. Discussion

4.1 Microstructural Evolution via AFSD Process

In the present work, the bottom half of the build deposited via AFSD was considered for the SCC studies due to the uniform hardness distribution shown in Fig. 3(b). The average grain size measured after the deposition process was 3.81 ± 2 µm. To determine if grain refinement was achieved, the obtained grain size was compared to grain sizes reported in the literature for wrought AA7075 rolled plate. Ahmed et al. (Ref 40), Jha et al. (Ref 41), and Venkit et al. (Ref 42) reported grain sizes of 40 µm, 34 µm, and 20.3 µm, respectively, in the rolling direction of the wrought AA7075. Comparing these literature values to the grain size of 3.81 ± 2 µm measured in the current study, it can be concluded that significant grain refinement was

achieved through the deposition process. Researchers have reported similar grain refinement for AA7075 processed using the AFSD or other friction stir-based technologies (Ref 17, 21, 25). Rokni et al. (Ref 43) reported similar grain refinement when they deposited AA7075 via a backward extrusion process. During the AFSD process, there is global plastic deformation and a high material strain rate, creating conducive conditions for dynamic recrystallization, hence grain refinement (Ref 4, 17, 43). The recrystallization process is controlled by the formation of dislocations and their accumulation at the grain boundaries, subgrains, and particle interfaces during the deformation process. This leads to the nucleation of defect-free grains (Ref 4, 44). The low fraction of LAGB observed from the misorientation distribution indicates a reduction in the dislocation density within grains, owing to the nucleation of defect-free grains during dynamic recrystallization in the AFSD process (Ref 17).

From texture analysis, there is evidence of shear texture from the pole figures and ODF plots (Fig. 5). Shear textures are often seen in materials that undergo deformation processes involving large strains (Ref 7, 37, 45) involving shear deformation. This large strain can be observed via friction stir welding (Ref 45), wear (Ref 46), torsion-based deformation processes (Ref 47), and AFSD (Ref 2, 7, 21, 48). The as-processed AA7075 exhibits A type, A^*, Weak Goss, and Brass type textures. From the literature, the resulting texture in the stir zone of friction stir welding (FSW) of aluminum alloys is the ideal texture (A, A*, C, and B) and Cube and Goss recrystallization components (Ref 49-53). In the present work, the C component was not present, while the Copper component was identified. The increase in temperature during torsional loading at large strains might have caused the C texture component to be substituted by the A and A* texture components, and the Weak Goss and Brass components are due to axial compression (Ref 47, 54). The specimens used in the experiments were taken from the bottom half of the build, which experienced repeated thermal cycles and axial compressive forces during deposition. Hence, although weak, the texture components present are in good agreement. Li et al. (Ref 55) suggest that Goss and Brass textures inhibit crack propagation, which in turn reduces the progression of stress corrosion cracks.

The TEM results (Fig. 6 and 7) informed the presence of intermetallic particles along the grain boundary and within intragranular regions. The intermetallic particles in as-processed conditions are coarse precipitates that may have formed and coarsened due to the repeated thermal cycle experienced by the bottom half of the AA7075 build used in the present study. Similar effect on intermetallic particles have been reported by Avery et al. (Ref 17). The intermetallic particles in as-processed conditions are enriched with Zn, Mg, and Cu, most likely η precipitates with a high concentration of Cu (typically observed in η phase precipitated dynamically). From the STEM-HAADF image, there were no η ($MgZn_2$) and η' ($MgZn_2$) strengthening precipitates present in the as-processed condition in the layers investigated.

In the T73 condition, coarse and numerous fine precipitates were observed. According to Orłowska et al. (Ref 56), the fine precipitates are the hardening precipitates formed during heat treatment. These fine precipitates might be GP(II) precipitates formed between 100 and 125 °C (Ref 57). In the present study, the first step of the T73 heat treatment was done at 107 °C, which falls within the temperature range to achieve GP(II). The coarse precipitates were mainly located along the grain boundary, with some interspersed between fine precipitates within the grains. The coarse precipitates likely precipitated during AFSD and were not fully dissolved during solutionizing as part of the T73 heat treatment. The T73 heat treatment caused the depletion of Cu from the remnant coarse η precipitates. In T73 condition, the EDS analysis showed that these coarse precipitates were only enriched in Zn and Mg. This observation agrees with studies done by Xiong et al. (Ref 58) on AA7075-T73. Although this study does not specifically address each particle type's role, Park and Ardell (Ref 59) compared the microstructures at the grain boundaries of Al-7075-T651 that underwent retrogression and re-aging, and T73 heat treatments. They analyzed the area fraction of particles on the grain boundaries (A_A), particle volume fraction (V_A), and the number of particles per unit area (N_A). Their findings indicated that the velocity of stress corrosion crack growth tends to decrease with higher A_A and V_A and lower N_A, improving resistance to SCC. However, particle chemistry also significantly impacts SCC resistance or susceptibility (Ref 60).

4.2 Microstructure-Mechanical Properties Relationship

Previous studies on as-deposited AFSD AA2219, AFSD AA6061, and AFSD Al-Li alloy by Rivera et al. (Ref 21), Phillips et al. (Ref 25), and Li et al. (Ref 23) did not observe such hardness gradient in their deposition. However, the height of their deposition did not exceed 10 mm. As reported by Mason et al. (Ref 54), repeated thermal cycling during the additive process causes the over-aging of the precipitates within the initial layers of the build when the height exceeds 10 mm. The as-deposited build showed degradation in hardness in this study (Fig. 3b). In the current study, the build deposition height was 50.8 mm, which subjects the initial layers of the build to repeated thermal cycling during deposition, causing the hardness to decrease significantly.

According to the literature, post-deposition heat treatment of AFSD aluminum alloys allows the previously dissolved precipitates to nucleate and grow once more, enabling improvements to the mechanical strength of the material (Ref 23, 26, 34, 61). The YS and UTS of the as-processed air-tested samples at 10^{-6} s^{-1} showed a reduction compared to the T73 counterpart (Table 2). During the AFSD process, the feedstock material with precipitates experiences microstructural changes where some precipitates dissolve, some dissolve and reprecipitate, and others overage (Ref 17, 23, 26). This agrees with the low hardness exhibited by the fourteenth layer from the top. Notably, the as-processed AA7075 demonstrated an increase in elongation to failure compared to the T73 condition of the AFSD AA7075 in the air environment. This is due to a lack of hindrance to dislocation motion by coarse precipitates (Ref 16, 26). Other researchers suggest that the refined grain structure obtained after deposition allows for accommodating more dislocations during tensile loading (Ref 20, 23).

Moreover, the thermomechanical process of AFSD promotes the recovery of the deformed structures and the annihilation of the dislocation, hence, the improvement in ductility (Ref 25). From the discussion of the TEM results of the as-processed condition, no η' ($MgZn_2$) hardening precipitates are present in the material. On the other hand, fine strengthening precipitates are re-precipitated during T73 heat treatment (Zn and Mg-rich). These precipitates obstruct dislocation movement, hence improving the strength but decreasing ductility.

4.3 Susceptibility of AFSD AA7075 to SCC

For this study, the SCC behavior of AFSD AA7075 was investigated through SSRT testing of smooth tensile coupons in a 3.5 wt.% NaCl solution at room temperature. The quantitative measurements obtained from the SSRT test technique can serve as indicators of the relative susceptibility of the material to SCC (Ref 62-65). The elongation of the tensile test specimens subjected to an air environment increased marginally with a decrease in strain rate, according to the results of the SSRT test (Fig. 8 and 9a). It is generally perceived that the plastic deformation of metallic materials displays a positive strain rate sensitivity to an increase in elongation with a decrease in strain rate (Ref 38, 66).

The SSRT test results for the tensile specimens tested in a salt environment exhibited a markedly different trend compared to the behavior observed for specimens tested in air. The elongation values exhibited a decreasing trend as the strain rate was reduced in the salt environment. Specifically, the specimen tested at the strain rate of 10^{-6} s^{-1} demonstrated the highest susceptibility to SCC within the range of strain rates investigated in this study. This is likely a consequence of the balance between the rate of crack propagation and the competing process of crack repassivation (Ref 38). In the salt solution, at relatively lower strain rates, the corrosive ions present in the solution are afforded sufficient time to interact with the as-processed AA7075. This extended interaction time enables the initiation of electrochemical reactions under loading, bringing about SCC, which manifests as a reduction in elongation.

The local strain at the crack tip varies microscopically with changes in the imposed strain rate, influencing the crack propagation. The local crack tip strain increases with the imposed strain and vice versa (Ref 67). The fracture tip is capable of maintaining a sharp geometry at high strain rates. Nevertheless, the fracture may be arrested and adopt a blunt shape if the repassivation rate at the crack tip surpasses the local crack tip strain rate during a slow strain rate test, irrespective of the environment. On the other hand, if the repassivation rate does not exceed the crack tip strain rate, the cracks will continue to propagate until the material fails. According to the literature, when the strain rates are relatively higher, the crack propagation occurs rapidly, and there is not sufficient time for the material to interact with the environment. Hence, the crack propagation rate is not affected by environmental factors (Ref 68-70). This indicates that the strain rate at the crack tip is influenced not solely by the overall imposed strain rate but also depends in part on the geometry of the crack tip, the local repassivation rate, and the immediate environment surrounding the crack tip.

For the SSRT test of as-processed AA7075 carried out at 10^{-6} s^{-1}, the SCC susceptibility was severe compared to tests carried out at 10^{-4} s^{-1} and 10^{-5} s^{-1}. The hypothesis is that the crack tip strain rate at 10^{-6} s^{-1} was significantly reduced, resulting in a cyclic process of passivation film dissolution, crack nucleation, crack propagation, and crack arrest. This was because the corrosive ions interacted with the as-processed AA7075 in a manner that was suggested by the slip-step dissolution (Ref 68, 69). At lower strain rates, the corrosive ions in the environment can interact with and dissolve the crack surfaces, affecting the geometry of the crack tip and causing it to become blunt. This can ultimately lead to the crack arresting as a passivation film form (Ref 67, 71). The cracks can resume propagation when the protective passive film re-fractures, which can occur due to an increase in the strain rate at the crack tip or a decrease in the repassivation rate. This change can be due to the local change in the chemistry of the solution around the crack tip (Ref 67). The hydrolysis reaction occurring at the crack tip can lead to the formation of an acidic solution in the localized environment, particularly when the diffusion of ions is restricted.

The passivation film can be disrupted by the reduction in pH and the subsequent increase in chloride ion concentration. This, in turn, decreases the repassivation rate, allowing more time for further dissolution of the material (Ref 68). The crack tip potential is also affected by the change in pH at the crack tip. According to Scully (Ref 68), SCC usually occurs in a material with passivation ability and in a region slightly below the pitting potential. The transition in the crack tip potential from a passivation domain to a cracking domain is caused by the increase in chloride ion concentration. The increase in chloride ion concentration is due to the decrease in pH. Aluminum alloys are known to have thin passivation films (Ref 72). Given the narrow repassivation region possessed by aluminum alloys, a probable shift in the crack potential during the SSRT test could provide a suitable condition for SCC, as a result, causing the propagation of cracks at lower strain rates. Repassivation of the crack potentially lowers the susceptibility to SCC by bringing the local pH closer to that of the bulk solution (Ref 70). From the preceding discussion, it is postulated that a decrease in strain rate has impacted the crack tip local chemistry, thus affecting the pH, the potential in the crack vicinity, and the repassivation rate. This ultimately leads to the SCC behavior shown by the as-processed AA7075.

4.4 SCC Mechanism: Comparison between As-processed and T73 Conditions

SCC can be described as the interplay between tensile stress, a chemically reactive environment, and a susceptible metal material together to facilitate the initiation and subsequent propagation of cracks (Ref 73. Varying heat treatment processes can lead to differences in the corrosion levels as well as the extent of material loss from the surface. Hence, it is hypothesized that the SCC is initiated by the grain boundary precipitates (GBPs) (Ref 74). During AFSD deposition, the initial layers are subjected to repeated thermal exposure. It can be thought that the initial layers undergo pseudo-heat treatment.

The study revealed the SCC behavior of tensile specimens subjected to T73 heat treatment. Grain boundary precipitates are formed during the heat treatment of aluminum alloys (Ref 75-77). The size, spacing, and elemental distribution of the precipitates that form along the grain boundaries significantly influence the SCC resistance (Ref 74). From the literature, Cu content in the grain boundary precipitates contributes to the SCC resistance (Ref 78-80). As shown in Fig. 7(d), Cu was distributed homogeneously in the matrix and grain boundary precipitates of the T73 condition compared to the as-processed condition where Cu is distributed in the precipitates within the grains and grain boundaries (Fig. 6d). The depletion of copper in the matrix creates a significant potential difference between the Cu-rich precipitates at the grain boundaries and the adjacent matrix. The matrix surrounding the Cu-rich precipitates can undergo selective corrosion, leading to the electrochemical separation of the precipitates from the alloy. This condition may lead to the initiation of SCC at the grain boundaries (Ref 60). This means a larger potential difference between the matrix and

the grain boundary precipitate is observed in the as-processed condition than in the T73 condition. This can promote SCC. Even though the large precipitate size improves SCC resistance, the GBPs in the as-processed condition of AA7075 become cathodic due to the Cu element present (Fig 6d). The absence of Cu in the matrix makes the Al matrix anodic, causing a potential difference between the grain boundary precipitate and the matrix. Hence, cracks will propagate easily through the matrix when initiated during SCC. These hypotheses agree with the higher SCC susceptibility index obtained by the as-processed condition than in the T73 heat-treated condition. Note that the retrogression and re-aging (RRA) heat treatment is another potential post-processing method that may further improve SCC resistance in AFSD-processed AA7075, as it has been shown to enhance SCC resistance in conventional AA7075-T6 (Ref 81).

5. Conclusions

i. The SCC behavior of as-processed AA7075 by AFSD in 3.5 wt.% NaCl solution by SSRT testing at 10^{-4} s^{-1}, 10^{-5} s^{-1}, and 10^{-6} s^{-1} initial strain rates was studied. The AFSD AA7075 alloy in T73 heat-treated condition was also subjected to SCC studies at 10^{-6} s^{-1} in the same environment. The tensile test specimens for the as-processed AA7075 and the T73 heat-treated AFSD were taken from the bottom half of the build deposition for SCC studies due to the uniform hardness distribution.

ii. The susceptibility of the as-processed tensile specimen to SCC was highest at 10^{-6} s^{-1}, with 10^{-4} s^{-1} and 10^{-5} s^{-1} showing almost the same susceptibility index. The T73 tensile specimen showed a lower susceptibility index compared to the as-processed tensile specimen tested at 10^{-6} s^{-1} strain rate.

iii. The EBSD, SEM, and TEM characterization techniques were used to understand the SCC behavior of AFSD AA7075. The as-deposited AA7075 had mostly equiaxed grains with a low fraction of low angle grain boundaries from the EBSD analysis. The fractographic study using the SEM showed that the as-processed tensile specimens exhibited shallow dimpled fractured features compared to the T73 tensile specimens fractured under the in salt solution environment.

iv From the TEM analysis, the T73 heat-treated condition of AFSD had Cu uniformly distributed within the grain and grain boundaries, while the as-processed condition had Cu distributed only in the precipitates within the grains. Most likely, the Al matrix of the as-processed condition was anodic due to Cu depletion compared to the second-phase particles enriched with Cu. The susceptibility of as-processed AA7075 via AFSD to SCC was likely due to the larger potential difference between the precipitate and the Al matrix. Hence, anodic dissolution occurs, allowing cracks to nucleate at sites with large potential differences.

6. Recommendations for Future Research

The findings reported in this study highlight key research directions for enhancing SCC resistance in solid-state processed aluminum alloys. The improved SCC resistance in T73 heat-treated AA7075 suggests that further investigation of post-processing treatments, such as retrogression and re-aging, may optimize corrosion resistance in AFSD and similar processes. The role of microstructural features, especially Cu distribution and grain boundary characteristics, also suggests that controlling these factors could reduce SCC susceptibility, informing future alloy design. Additionally, insights into anodic behavior and precipitate-matrix interactions encourage further mechanistic research to identify microstructural traits affecting SCC. Finally, testing in application-specific environments, such as marine or aerospace conditions, could refine processing techniques for enhanced durability in corrosion-prone settings. These directions can help advance solid-state manufacturing for critical applications.

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